VYAPAARSAATHI

Bolo, aur hisaab ho jaye. | Voice-First Bookkeeping & Business Assistant

Official Presentation, Architecture Walkthrough & Examiner Viva Defense Guide

Candidate: Atharv Ravindra Gole

Stack: FastAPI | Groq Whisper & LLM | MongoDB | React

1. 30-Second Elevator Pitch (Opening Hook for Presentation)

VERBATIM PRESENTATION HOOK

"Good morning, respected examiners. Millions of small shopkeepers and street vendors across India conduct their day-to-day business naturally in spoken Hindi, Marathi, and Hinglish. Traditional accounting software fails them because it expects complex desktop forms, English accounting jargon, and tedious manual typing.

VyapaarSaathi solves this with a simple promise: 'Bolo, aur hisaab ho jaye.' A vendor sends a natural WhatsApp voice note or text. Our platform cleans street noise, transcribes speech with Groq Whisper, extracts business transactions via LLMs, maintains real-time inventory, tracks customer credit (Udhari), and delivers an explainable morning business plan at 09:00 AM IST.

Crucial Engineering Principle: AI understands the natural spoken language, but 100% deterministic Python code enforces mathematical calculations, inventory balances, and ledger integrity."

2. End-to-End System Architecture & Data Pipeline

THE 7-STAGE EXECUTION PIPELINE

1. WhatsApp Ingestion: Vendor sends voice note (.ogg Opus) or text via WhatsApp to Meta Cloud API webhook.
2. Audio Preprocessing: FastAPI downloads binary audio and runs an FFmpeg DSP filter chain (highpass, lowpass, afftdn, loudnorm) to strip traffic rumble and normalize loudness.
3. Transcription: Preprocessed 16kHz mono audio is transcribed by Groq Whisper Large-v3-Turbo in ~350ms, with prompt guidance to retain Hindi/Marathi/Hinglish vocabulary.
4. Phonetic & Spoken Normalization: Spoken numbers ('paanso' -> '500', 'bees' -> '20') and phonetic terms ('uadhar' -> 'udhari') are standardized using regex.
5. Structured Extraction: Groq LLM (Mixtral / LLaMA) parses transcripts into strict JSON matching Pydantic schemas (10 distinct intents: Sales, Expenses, Stock, Udhari, Reports).
6. Conversational State Machine: If required data is missing (e.g. item sold without quantity), the engine triggers an in-memory pending state, asks a Hinglish counter-question, and returns TTS voice.
7. Deterministic Ledger & Inventory: Mathematical totals (qty x price), stock deductions, and customer debt updates are computed in Python and persisted in MongoDB Atlas.

FFmpeg Noise Suppression Command Chain:

```
ffmpeg -y -i input.ogg -ac 1 -ar 16000 \
  -af highpass=f=100,lowpass=f=7000,afftdn=nr=12:nf=-35,loudnorm=I=-16:TP=-1.5:LRA=11 \
  normalized.mp3
```

3. Core Functional Modules

- Voice Ledger (voice_ledger.py)

Handles Meta webhook payloads, audio conversion, transcription, multi-intent extraction, Pydantic validation, and formatted bilingual WhatsApp confirmation messages with daily profit breakdown.

- Conversation Engine (conversation_engine.py)

Manages multi-turn voice dialogs. Evaluates extracted data against live catalog. If incomplete (NEEDS_QUANTITY, NEEDS_AMOUNT, NEW_ITEM), pauses execution, synthesizes counter-question audio via gTTS, and merges follow-up replies.

- Catalog & Inventory (inventory.py)

Real-time stock decrement upon sales. Automatic classification of stock health: Healthy (>20%), Low Stock (<=20%), or Stockout (0 left). Includes 1-click morning quota reset functionality.

- Customer Credit / Udhari (udhari.py)

Double-entry customer credit management. Records credit (UDHARI_ADD), partial payments, and total clearances (clear_all=True). Includes 24-hr debounced WhatsApp reminders and FPDF2 statement export.

- Business Autopilot (autopilot.py & scheduler.py)

Deterministic advisory engine scheduled via APScheduler at 09:00 AM IST. Calculates 14-day rolling income/expenses, identifies restock quantities, and generates targeted debt recovery alerts.

- Web & Admin Portals (frontend/src/)

React 18 + Vite + Tailwind application. Features vendor login via WhatsApp OTP with anti-brute-force rate limiting, live ledger/stock dashboards, and an Admin panel with audit logs and broadcast tools.

4. Live Presentation Demo Playbook (Step-by-Step)

Step 1: Standard Sale & Expense Recording (Instant Ledger)

Speak / Type: 'Aaj 25 chai aur 15 samosa beche, kul 550 rupaye mile, aur doodh par 180 kharch hua.'

Observe: System detects income and expense, decrements stock (-25 chai, -15 samosa), records Rs. 180 expense, and replies on WhatsApp showing net profit of Rs. 370.

Step 2: Incomplete Voice Entry (Conversation Engine & Audio Follow-up)

Speak / Type: 'Bhaiya samosa bech diya.' (Omit quantity and amount)

Observe: Status becomes NEEDS_QUANTITY. WhatsApp bot replies with voice note + text: 'Aapne samosa beche - kitne?'

Vendor Replies: '20 piece'.

Observe: System auto-multiplies 20 x Rs. 15 (catalog price), updates stock (-20), and saves Rs. 300 income!

Step 3: Customer Credit (Udhari Transaction & Settlement)

Speak / Type: 'Ramesh ko 250 rupaye udhari diya.'

Observe: Customer 'Ramesh' is credited with Rs. 250 debt balance.

Speak / Type: 'Ramesh ne 100 rupaye wapas diye.'

Observe: Balance decreases to Rs. 150. Show the 'Download Udhari PDF' button on web dashboard.

Step 4: Real-Time Stock Status Inquiry
Speak / Type: 'Mera stock dikhao.'
Observe: WhatsApp replies immediately with complete item inventory and flags any low-stock items.

Step 5: Web Dashboard & Admin Portal
Navigate to Web UI: Show live Ledger entries, Inventory health badges, AI Insights charts (Recharts), and Admin Portal with audit trail.

5. Comprehensive Examiner Viva Question Bank (Top 18 Q&As)

[AI & SPEECH] Q1: Why did you use Groq Whisper and LLM instead of Google Cloud STT or local models?

Three reasons: (1) Latency: Groq LPUs run Whisper in ~350ms and LLM extraction in ~400ms, beating the 1.5s webhook deadline. (2) Indic Code-Mixing: Whisper Large-v3 handles mixed Hindi/Marathi/Hinglish acoustics seamlessly. (3) Local Fallback: If cloud fails, the backend automatically falls back to an in-memory OpenAI Whisper base model.

[AI & SPEECH] Q2: How do you handle heavy background street noise (horns, market chatter)?

Every voice note passes through an FFmpeg DSP filter pipeline before Whisper: (1) highpass=100Hz cuts low vehicle engine rumble, (2) lowpass=7000Hz removes high-frequency whine, (3) afftdn dynamically estimates and cancels ambient noise, and (4) loudnorm standardizes speech volume to EBU R128 (-16 LUFS).

[AI & SPEECH] Q3: How do you guarantee LLMs won't hallucinate numbers and corrupt the ledger?

The LLM is strictly used as an NLP extraction parser, NEVER as a calculator. Outputs are constrained to a strict Pydantic JSON schema. All mathematical multiplications (qty x price), running balances, and daily profit figures are computed deterministically in Python code. If price is omitted, it is looked up from MongoDB inventory.

[AI & SPEECH] Q4: How do you resolve phonetic Whisper misspellings and spoken Hindi numbers?

In services/voice_ledger.py, normalize_spoken_numbers() executes regex transformations prior to LLM processing: converts colloquial terms ('paanso' -> '500', 'uadhar' -> 'udhari', 'rubia' -> 'rupya') and parses compound spoken units ('bees sau' -> 2000, 'paanch hazaar' -> 5000) in both Latin and Devanagari scripts.

[STATE MACHINE] Q5: What happens if a vendor sends an incomplete voice note like 'Maine samosa beche'?

The ConversationEngine analyzes the input against the catalog. Finding samosa exists at Rs. 15 but quantity is missing, it sets state to NEEDS_QUANTITY. It stores pending state in memory, asks a natural Hinglish counter-question ('Aapne samosa beche - kitne?'), and generates TTS audio. When the vendor replies, it merges the data and saves.

[STATE MACHINE] Q6: Why is pending conversation state stored in memory rather than a database?

Follow-up replies happen within 5-15 seconds. In-memory storage gives microsecond lookups with zero database I/O latency. For a distributed cluster deployment, this can be swapped with Redis using a 5-minute Time-To-Live (TTL).

[BUSINESS LOGIC] Q7: How does the Udhari ledger prevent harassment or duplicate reminders?

Udhari customer reminders sent via WhatsApp have an automated 24-hour debounce cooldown in MongoDB. If a vendor tries to spam reminders within 24 hours, the backend blocks the dispatch to protect customer relations.

[ARCHITECTURE] Q8: Why did you choose MongoDB instead of a relational database like PostgreSQL?

Small vendor businesses have heterogeneous schemas (chai per cup vs clothes per meter). Document-oriented MongoDB stores variable catalog metadata without schema migration locks. Furthermore, every collection (ledger, inventory, udhari) is naturally sharded and indexed on { phone: 1, created_at: -1 } for fast tenant-isolated queries.

[SECURITY] Q9: How do you ensure multi-tenant vendor isolation so Vendor A cannot view Vendor B's data?

Every database query strictly enforces a phone boundary filter { phone: normalized_phone }. Phone variants (+91, 91, whatsapp JID) are normalized via helper functions. On the web dashboard, all API routes are guarded by JWT tokens containing the verified vendor phone, verified via FastAPI dependencies.

[ARCHITECTURE] Q10: How do you handle Meta's 15-second webhook timeout requirement?

When incoming events hit POST /webhook, FastAPI verifies the payload and returns HTTP 200 immediately, dispatching audio downloading, FFmpeg filtering, Whisper STT, and LLM extraction to FastAPI BackgroundTasks. This prevents webhook retries and duplicate processing.

[WHATSAPP API] Q11: How does the WhatsApp 24-hour customer service window affect proactive messaging?

When a vendor contacts the bot, a 24-hour free-form session opens. For messages initiated outside 24 hours (like the 09:00 AM Business Autopilot or inactive vendor nudges), Meta requires pre-approved HSM utility templates. Our architecture is ready with send_whatsapp_template().

[AUTOPILOT] Q12: How is the Business Autopilot calculated? Is it generative AI or deterministic rules?

Autopilot uses deterministic business heuristics, NOT generative speculation. It calculates 14-day rolling income/expenses, flags items whose current stock is <= 20% of daily quota, computes exact restock quantities, and prioritizes overdue debts. Dispatches are scheduled by APScheduler at 09:00 AM IST with duplicate-delivery prevention.

[SECURITY] Q13: How is security handled across authentication and the admin portal?

Admin passwords use salted bcrypt hashing (work factor 12). Vendor web sessions use cryptographically generated 4-digit OTPs delivered via WhatsApp with 10-minute expiry and anti-brute-force lockout. Sessions are secured with HMAC-SHA256 JWTs. All admin actions are written to admin_audit_logs with client IP and timestamps.

[SYSTEM DESIGN] Q14: What are the 10 supported voice intents in VyapaarSaathi?

ADD_ENTRY (income/expense), STOCK_UPDATE (adjust stock), PRICE_UPDATE (change unit price), GET_REPORT (PnL summary), GET_INVENTORY (stock health status), DELETE_ITEM (soft delete catalog item), UDHARI_ADD (give credit), UDHARI_PAYMENT (receive debt payment), UDHARI_QUERY (check balance), and UDHARI_REPORT (full debt list).

6. Final Presentation Day Checklist & Emergency Tips

MORNING PRESENTATION CHECKLIST

- System Pre-flight: Ensure MongoDB Atlas is connected, Groq API key is active, and WhatsApp webhook is verified.
- Local Fallback: If internet is unstable, mention that local Whisper (base model) and static mock endpoints are built in.
- Confident Opening: Begin directly with the 30-Second Elevator Pitch. Don't start with boring slides on history of accounting.
- Emphasize Safety: Whenever asked about AI risks, repeat: 'AI handles transcription and parsing; deterministic Python code handles accounting math.'
- Showcase Real Impact: Highlight that this directly empowers non-English-speaking, non-tech-savvy micro-entrepreneurs across Bharat.